

Control Flow

This project is split into a `synth.py` and `synth_helpers.py` file. In the `synth` file a prompt is made that asks the user for an input path to a midi file that then gets transformed into a list of tuples representing note data. The command prompt then asks for a series of parameters (if the user doesn't choose a preset value) that's saved to a dictionary. This param dictionary is used in conjunction with the note data to iterate note by note using the `synth_helpers` functions to do dsp synthesis to create sounds with the correct parameters and pitch at the right time. This buffer is then written to a file in the directory.

Example Sounds

- Lethal Lava Land
 - Uses the "Sitar" preset to emulate the iconic intro to Lethal Lava Land from Super Mario 64.
- Carol of the Bells
 - Use the "Cold Bells" preset to make a cover of Carol of the Bells using FM synthesis.
- 1812 Overture
 - Used fm synthesis on a sawtooth wave with an index of 3 and a modulation of 2 with fast attack and release times to emulate the tambre of a trumpet.